

■ 2■■ CONVERT PROJECT INTO A WEB APP (STREAMLIT)

Now let's convert your ML model into a working web application.

■ CreditWise Loan Approval – Web App

■ Technology Used

Streamlit (Frontend)

Scikit-learn (Model)

Joblib (Model loading)

Python

■ Project Structure

CreditWise_WebApp/

■

■■■ app.py

■■■ loan_approval_model.pkl

■■■ loan_approval_data.csv

■■■ requirements.txt

■ Step 1: Install Required Libraries

```
pip install streamlit scikit-learn pandas joblib
```

■ Step 2: Save Your Trained Model

```
import joblib
```

```
joblib.dump(model, "loan_approval_model.pkl")
```

■ Step 3: Create app.py

```
import streamlit as st
```

```
import joblib
```

```
import numpy as np
```

```
# Load model
```

```
model = joblib.load("loan_approval_model.pkl")
```

```
st.title("■ CreditWise Loan Approval System")
```

```
st.write("Enter applicant details to predict loan approval")
```

```
income = st.number_input("Applicant Income", min_value=0)
```

```
credit_score = st.number_input("Credit Score", min_value=300, max_value=900)
```

```
loan_amount = st.number_input("Loan Amount", min_value=0)
```

```
dti = st.slider("Debt to Income Ratio", 0.0, 1.0)
```

```
if st.button("Check Loan Status"):
```

```
data = np.array([[income, credit_score, loan_amount, dti]])
```

```
prediction = model.predict(data)
```

```
if prediction[0] == 1:
```

```
st.success("■ Loan Approved")
```

```
else:
```

```
st.error("■ Loan Rejected")
```

■ Step 4: Run the Web App

```
streamlit run app.py
```

➡■ Opens automatically in browser

➡■ Fully interactive loan approval system